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DIVAS: an R package for identifying shared and individual variations of multiomics data

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Abstract

Motivation: Multiomics data integration aims to identify biological patterns shared across different molecular modalities. Traditional methods focus on detecting either jointly shared variation (across all modalities) or individual variation (unique to single modalities), but overlook partially shared variation, shared by subsets of modalities. This limitation is critical because many biological mechanisms manifest across only some, not all, molecular modalities. Systematically identifying these partially shared patterns is essential for comprehensive understanding of complex biological systems.

Results: We developed an open-source R package that implements DIVAS, a computational framework for systematically identifying jointly shared, partially shared and individual variations across multiple data types. DIVAS employs angle-based subspace analysis with rigorous statistical inference through rotational bootstrap. Unlike existing methods, DIVAS hierarchically searches through all possible combinations of data modalities, providing complete decomposition of multiomics data into interpretable components (modes of variation) with both scores and loadings. Application to multimodal COVID-19 data demonstrates DIVAS's ability to reveal biologically meaningful partially shared immune and metabolic dysregulation patterns that underpins disease severity and would be missed by conventional integration approaches.

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Availability and implementation: DIVAS is available as an R package on GitHub (`https://github.com/ByronSyun/DIVAS_Develop`). A step-by-step vignette is available on GitHub (`https://byronsyun.github.io/DIVAS_COVID19_CaseStudy/`).

1 Introduction

The integration of multiple omics data types has become essential for understanding complex biological systems. Modern high-throughput technologies enable simultaneous measurement of diverse molecular layers, including genomics, transcriptomics, proteomics and metabolomics, from the same biological samples. These multiomics datasets offer complementary views of biological processes, yet analysing them in a unified framework remains challenging. The key question in multiomics integration is: how do we systematically identify which biological signals are shared across all omics types (*jointly shared*), which are shared by only some types (*partially shared*) and which are unique to individual omics layers (*individual*)?

There exist many multivariate statistical approaches for multiomics integration. Most of these approaches are limited to identifying jointly shared variations, e.g. methods based on Canonical Correlation Analysis (CCA) or Partial Least Squares (PLS) identify lower-dimensional representations of multiomic data by maximising correlations between input data blocks. Widely used methods based on CCA or PLS include Regularised CCA (RCCA) [1], sparse RCCA [2] and Regularised Generalised CCA (RGCCA) [3] and DIABLO [4].

Joint and Individual Variation Explained (JIVE) [5] represented a significant advance by decomposing multiomics data into joint variation shared across all data blocks and individual variation unique to each block. A later generation, Angle-based JIVE (AJIVE) [6], improved upon JIVE by using principal angle analysis for faster joint structure selection and better theoretical justification through matrix perturbation theory.

Multi-Omics Factor Analysis (MOFA) [7] and its extension MOFA+ [8] started to explore partially shared variations, in addition to jointly shared and individual variations. MOFA and MOFA+ use a Bayesian group factor analysis framework to identify latent factors that explain variation within and across omics layers. MOFA+ extends MOFA by handling

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multiple groups of samples and providing better scalability. While MOFA and MOFA+ can identify factors with varying degrees of sharing across data modalities, they do not systematically enumerate all possible patterns of partial sharing, and the interpretation of which specific subset of blocks shares each factor can be ambiguous.

Recent methods started to focus on mining partially shared variations. Structural Learning and Integrative Decomposition (SLIDE) [9] models variation shared between specific subsets of data blocks using structured sparsity, determining block-specific sparsity patterns through bi-cross-validation. However, SLIDE faces challenges in the systematic detection of all possible patterns of partial sharing and it lacks rigorous statistical inference for the identified patterns. More recently, Yi et al. [10] proposed Hierarchical Nuclear Norm (HNN) penalisation that formulates partially shared variations through hierarchical levels and uses a convex optimisation approach avoiding scores-loadings factorisation. While HNN provides theoretical guarantees for identifiability and convex formulation, it requires careful tuning parameter selection via a computationally intensive cross-validation procedure. No computing package is available for SLIDE nor the HNN method, increasing the difficulty of their application.

Comparison of key features of the aforementioned methods are summarised in Table 1. The critical methodological gap lies in systematically identifying partially shared variation, representing biological signals shared by specific subsets of omics modalities. The ability to reliably identify partially shared variations is particularly important when there are 3 or more input modalities. Focusing on jointly shared variations may be problematic since there is no guarantee that the most interesting biological patterns will be captured by such variations. For instance, certain regulatory mechanisms may affect transcriptomics and proteomics but not metabolomics, while metabolic dysregulation may be captured by metabolomics and proteomics but not reflected at the transcriptional level.

Addressing this critical research gap we introduce data Integration Via Analysis of Subspaces (DIVAS), originally proposed by Prothero et al. [11] and implemented in the closed source MATLAB (see also Ackerman et al. [12] for an application in neuroimaging). We implemented DIVAS as an open source R package with additional visualisation tools and much improved user friendliness. Compared to existing methods, DIVAS enjoys the following

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advantages.

1. Systematic identification of jointly shared, partially shared and individual variations;
2. Rigorous inference procedures enabling statistical guarantee;
3. Implementation of broadly useful defaults, enabling completely data-driven integration without extensive manual parameter tuning.

Next we introduce the DIVAS methodology in some detail. DIVAS's capabilities are then demonstrated using a COVID-19 multiomic dataset.

Table 1: Comparison of multiomics integration methods. '✓' indicates that a method is capable of performing a specific task, while '✗' denotes that the method is not equipped to handle the task.

Method	**Jointly shared**	**Partially shared**	**Individual**	**Statistical inference**	**Package availability**
RCCA/RGCCA	✓	✗	✗	✗	CRAN [13]
DIABLO	✓	✗	Limited[a]	✗	Bioconductor [14]
JIVE	✓	✗	✓	Limited[b]	CRAN [15]
AJIVE	✓	✗	✓	Limited[c]	GitHub [16]
MOFA/MOFA+	✓	Limited[d]	✓	✓[e]	Bioconductor [17]
SLIDE	✓	✓	✓	Limited[f]	N/A
HNN	✓	✓	✓	Limited[g]	N/A
DIVAS	✓	✓	✓	✓	GitHub [18]

[a] Focus on joint variation; individual variation not explicitly modelled
[b] Permutation testing for rank selection only
[c] Angle bounds for joint vs individual separation, but not for rank selection
[d] Learns sparsity patterns but does not detect all partial sharing patterns; interpretation shared patterns can be ambiguous
[e] Bayesian inference on factors and loadings
[f] Cross-validation for model selection only
[g] Convex optimization with tuning parameter selection via bi-cross-validation

2 Methods

The input of DIVAS is a collection of K data blocks $X_1, \ldots, X_K$ from N samples, where each X_k is a $(d_k \times N)$ feature-by-sample matrix. These data blocks represent multiomic measurements from the same samples. The DIVAS algorithm, detailed in Prothero et al. [11], comprises the following two main steps:

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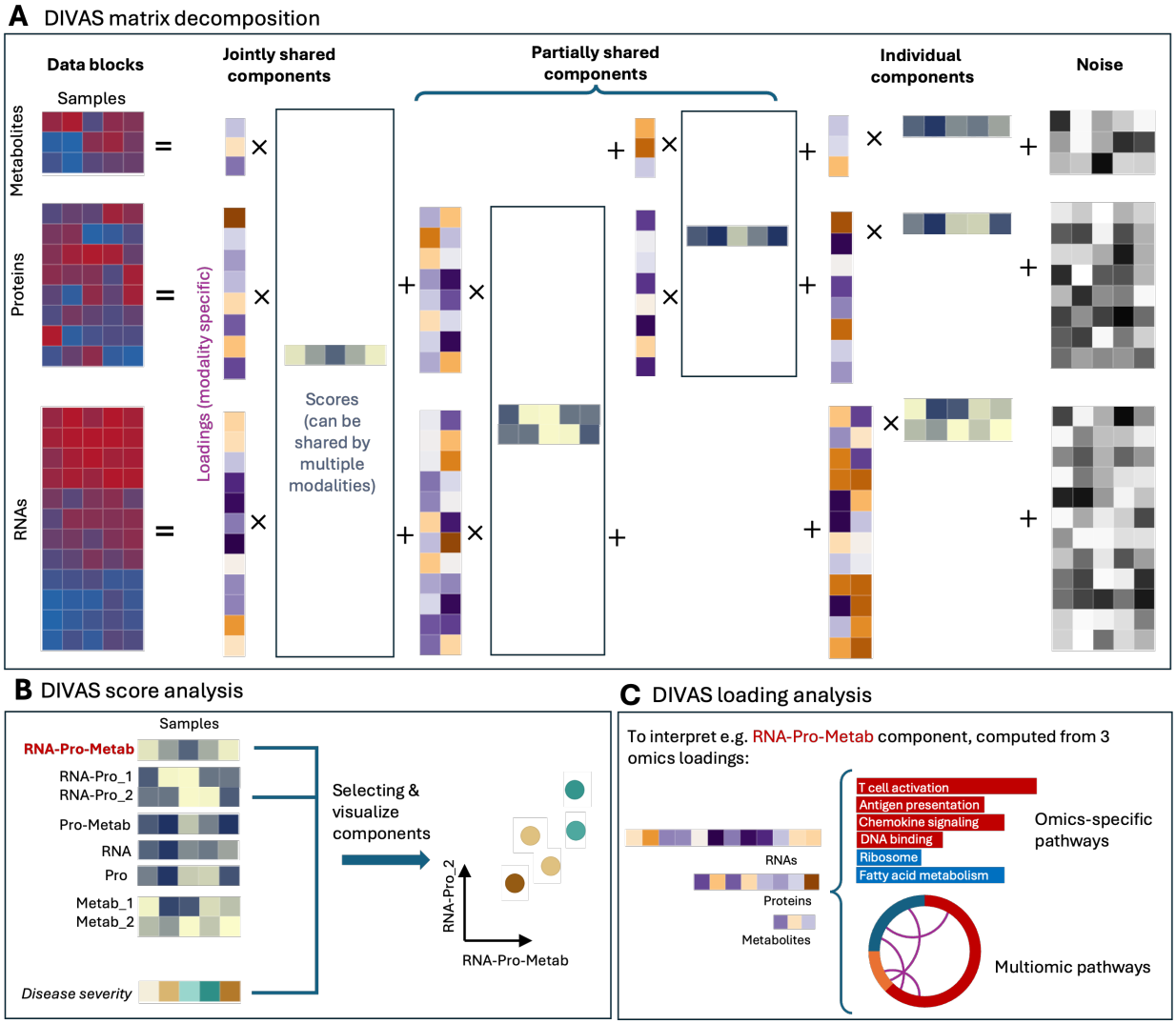

Fig 1: Schema of DIVAS. **A** DIVAS decomposes input data blocks into components representing jointly shared, partially shared and individual modes of variation. Here we present a toy example taking transcriptomics, proteomics and metabolomics as input. Each DIVAS component, representing a specific mode of variation, is defined by a score vector and one or more loading vectors (corresponding to one or more modalities). In the toy example, we found one 3-way jointly shared component (RNA–protein–metabolite), three 2-way partially shared components (two RNA–protein, one protein–metabolite) and four individual components (two RNA, one protein, one metabolite). **B** DIVAS scores are lower dimensional summaries of samples and can be used for visualisation. For example, we can select the scores most highly correlated with disease severity and visualise them using a scatter plot. **C** DIVAS loadings measure the contribution of individual omics features to scores. Whereas one DIVAS component only has one common score vector, it may have multiple loading vectors, corresponding a set of data blocks. By identifying pathways represented by important omics features, we can interpret the biological meanings of individual components.

1. Denoising individual data blocks;

2. Identifying shared and individual structures among denoised data blocks.

Both steps are fully data-driven and tuning free. The output of DIVAS are components

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explaining shared and individual variations, where each component has corresponding scores and loadings as in standard factor models. Next we present some more details of the DIVAS algorithm and downstream analyses based on the scores and loadings; see Fig 1 for a schema of DIVAS.

Denoising data blocks. We assume that each data block has an underlying low rank structure A_k, which is contaminated by observational noise E_k (see Fig 1A). That is, we assume $X_k = A_k + E_k$. To recover A_k from noisy observation X_k, we apply Principal Component Analysis (PCA) to X_k and take the first PCs. The number of PCs to use is determined by an optimal shrinkage method [19, 20] in a fully data-driven way.

Identifying shared and individual structures. We assume that the total variation in each A_k, the noiseless version of the kth data block, can be decomposed into jointly shared, partially shared and individual variations. That is, with $\boldsymbol{i}$ denoting a subset of the index set of all data blocks $\{1, \ldots, K\}$, we assume that

$$A_k = \sum_{\text{all the } \boldsymbol{i}\text{'s that contain } k} L_{\boldsymbol{i},k} S_{\boldsymbol{i}}^{\top},$$

where $S_{\boldsymbol{i}}$ is a $(r_{\boldsymbol{i}} \times N)$ score matrix, with $r_{\boldsymbol{i}}$ denoting the number of components shared by data blocks in $\boldsymbol{i}$, and where $L_{\boldsymbol{i},k}$ is a $(d_k \times r_{\boldsymbol{i}})$ loading matrix, linking the $r_{\boldsymbol{i}}$ components with the d_k features of the kth data block. That is, a DIVAS component shared by data blocks in $\boldsymbol{i}$ comprises *common* scores for all those data blocks, and K different loading vectors for individual data blocks; see Fig 1A for an example involving 3 data blocks.

The core of the DIVAS algorithm is a sophisticated convex-concave optimisation procedure [11]. Here we only present a sketch of the algorithm. DIVAS systematically searches for shared structures using a hierarchical approach, starting with the highest-order structures (shared among more data blocks) and progressively moving to lower-order structures (shared among fewer blocks or individual to single blocks). In the context of multiomics data, a shared structure represents a pattern of biological variation that captures how different omics variables co-vary across samples. Finding these structures boils down to identifying specific directions that point towards the most important patterns of variation.

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The DIVAS algorithm then uses an iterative approach to test each candidate direction to see if it genuinely represents shared biology by checking whether it stays close to the signal patterns in all included data blocks, whilst remaining sufficiently distant from excluded blocks. This process continues until no additional meaningful directions can be found for the current structure. The algorithm then moves on to search for the next type of shared pattern. The unique strength of this approach is that it automatically determines how many dimensions are needed to capture each type of biological relationship, providing a data-driven decomposition of complex multiomics datasets into interpretable shared and individual components.

Interpreting scores and loadings. The DIVAS decomposition yields two key outputs for each identified component: scores and loadings. Each component has a score vector representing a one-dimensional summary of how samples vary along a particular mode of variation (Fig 1A&B). These scores are shared across all data blocks involved in a given structure. For example, a jointly shared component across transcriptomics, proteomics and metabolomics has one common score vector, indicating a biological pattern across all three modalities.

Loadings link the shared scores to the specific features measured in each data block. While a component has only one score vector, it may have multiple modality-specific loading vectors (Fig 1A&C). Each loading vector indicates which features in that particular data block contribute to the shared variation captured by the component's score. For example, for a jointly shared component in Fig 1 C, the transcriptomics loading vector reveals which genes drive that biological pattern, while the corresponding proteomics and metabolomics loading vectors identify the associated proteins and metabolites. We can then take the top contributing omics features for pathway analyses (Fig 1C).

3 Results

Study design and data. We analysed a multiomic dataset from a cohort of mild to severe COVID-19 patients [37] to illustrate how DIVAS can streamline biological discoveries in

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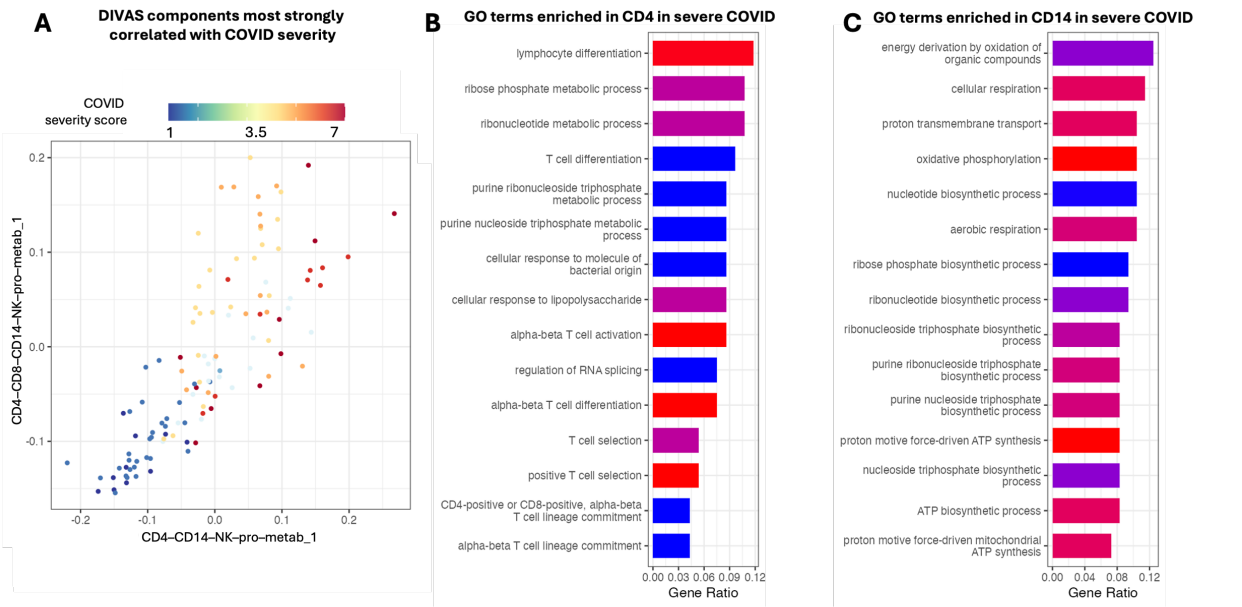

D

Metabolic patterns	Metabolic markers	Refs
Amino acid metabolism dysregulation	Cysteine, cystathionine, quinolinate, kynurenine	[21,22]
Lipid metabolism alterations	Carnitines, Linoleoyl compounds, lysophospholipids	[23–25]
Bile acid metabolism	Taurocholate, deoxycholic acid, glycocholate derivatives	[26,27]
Oxidative stress markers	Glutarate, sulfated compounds	[22,28]
Proteomic patterns	**Proteomic markers**	**Refs**
Cytokine storm	TNF receptor superfamily, interleukins and receptors, interferon lambda receptor 1	[29,30]
Immune cell activation	S100A12, CD79b, SLAM family	[31,32]
Interferon response	Interferon lambda receptor 1	[33]
Tissue damage	Matrix metallopeptidase, coiled-coil domain proteins, adhesion molecules, Gelsolin-like capping proteins	[34–36]

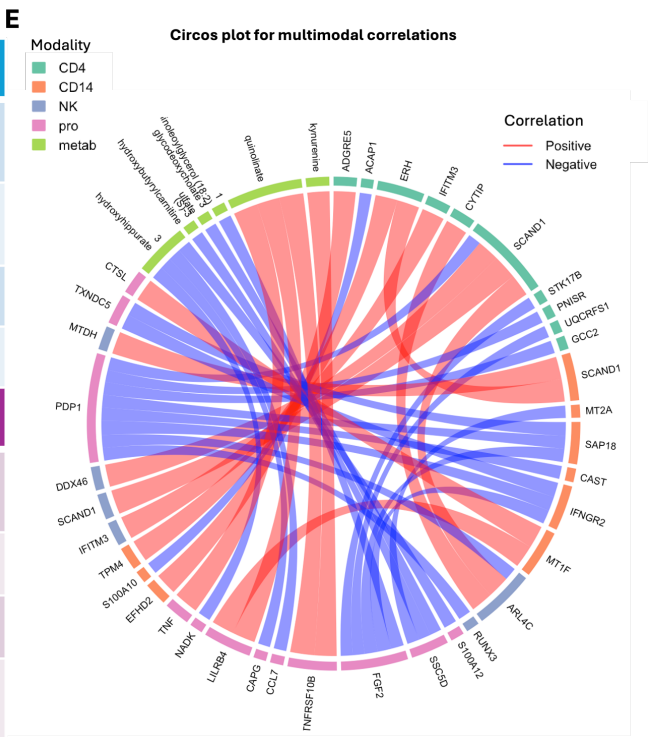

Fig 2: Uncovering multiomic pathways underpinning COVID-19 severity. **A** Visualising the scores of DIVAS components most strongly correlated with COVID severity. A component, e.g. CD4–CD14–NK–pro–metab_1, is named after the data blocks it represents. As there may be multiple components corresponding to a particular combination of data blocks, we also use an index to distinguish them. **B&C** Gene ontology (GO) terms enriched in CD4 T cell and CD14 monocyte gene expressions in severe COVID-19. **D** Metabolic and proteomic patterns observed in severe COVID-19 [21–36]. **E** Circos plot illustrating within- and cross-modal correlations, where the thickness of ribbon represents the absolute value of pairwise Spearman correlation coefficient between connected features.

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complex multiomic datasets. The study included 139 patients measured at two time points, T1 and T2, resulting in 265 patient samples. Measurements of multiple omics types were available for these patients, we selected scRNA-seq, bulk proteomics and bulk metabolomics as input to DIVAS. The number of patients with all three types of omics was 120; the total number of samples were therefore 240 (two time points). Since DIVAS assumes independence between samples for statistical inference purposes, we needed to avoid including repeated measurements of the same subjects to break the correlation structure. Hence we randomly sampled 60 individuals and took their T1 observations, combined with observations of the remaining 60 individuals at T2. As a result we had 120 samples in total, including observations from both time points but not including repeated measurements of the same patients.

To harmonise the cell-level scRNA-seq and sample-level proteomisc and metabolomics, we pseudo-bulked CD4 T, CD8 T, CD14 monocyte and natural killer (NK) cells from each sample. The four cell types were selected considering their important roles in COVID-19 immune response [37]. We then treated the gene expression (GEX) profiles of these 4 cell types, along with proteomics (pro) and metabolomics (metab), as 6 input data blocks to DIVAS. See Supplementary Materials Section S1 for a detailed account of the preprocessing and cell type annotation of scRNA-seq.

Biological insights. Applying DIVAS to the 6-modal COVID-19 data, we found over 90 statistically significant components, ranging from 6-way (jointly shared variation) down to 1-way (individual variation) (see Supplementary Fig S1). We then ranked all DIVAS component according to the correlation between their scores with the COVID-19 severity score, an ordinal variable ranging from 1 to 7. The top two most highly correlated components are visualised in Fig 2A (see Supplementary Fig S2 for the top five components).

DIVAS loadings allows us to rank the modality-specific features according to their contribution (loading values) to a component of interest. For CD4–CD14–NK–pro–metab_1, the component most strongly correlated with COVID-19 severity, we took CD4 GEX features most positively and negatively associated COVID-19 severity and conducted gene set overrepresentation analysis (see Supplementary Methods Section S2). The results marked drastic differences in phenotypes of CD4 T cells in severe versus mild COVID-19 samples.

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While mild COVID-19 patients showed a more coordinated immune response, marked by T cell and hematopoiesis regulations (Fig 2B), more severe COVID-19 patients showed highly activated but dysfunctional CD4 T cell state, marked by metabolic stress (Fig 2C). From the proteomics and metabolomics modalities, we found markers representing a series of immune and metabolic dysregulations, as summarised in Fig 2D. These findings confirmed the CD4 T cell dysregulation in severe COVID-19 reflected in CD4 GEX in Figs 2B&C, and provided additional insights from the perspectives of different omics types.

Finally, we identified the top features from all 5 modalities involved in the CD4–CD14–NK–pro–metab_1 component and examined their correlations (see Supplementary Methods Section S2). Interestingly, we identified correlation structures that helped explaining the immune dysregulation mentioned above. Namely multiple chemokines (e.g. CCL7), TNF receptors (e.g. TNFRSF10B) and metabolites (e.g. quinolinate) formed an interconnected network (Fig 2E), suggesting the cytokine storm that is commonly observed in severe COVID-19 immune response [29, 30, 33]. Another highlight was the multiple strong negative correlations identified across modalities (blue links in Fig 2E). For example, PDP1 protein was found to be negatively correlated with IFNGR2 gene in CD14 monocytes. This suggested a discordance in severe COVID-19 patients between myeloid activation, for which IFNGR2 is essential [38], and energy production via oxidative metabolism, promoted by PDP1 [39].

4 Conclusion

We introduced the DIVAS R package, implementing a framework for multiomics data integration that systematically identifies jointly shared, partially shared and individual variations across multiple input data blocks. DIVAS fills a critical gap in the multiomics integration toolkit by enabling statistically principled detection of partially shared biological patterns. This capability is essential because many biological mechanisms are reflected only on certain molecular modalities. Traditional methods that focus exclusively on fully shared or individual variations miss these important intermediate patterns.

The key innovations of DIVAS include: (1) hierarchical search through all possible combinations of data modalities using angle-based subspace analysis; (2) built-in statistical in-

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ference through principal angle analysis and rotational bootstrap; (3) simultaneous consideration of both sample and feature spaces for enhanced accuracy; and (4) effective handling of high-dimensional data through optimal shrinkage methods. Our analysis of multi-modal COVID-19 data demonstrates how DIVAS reveals biologically meaningful partially shared structures such as immune dysregulation patterns manifested in specific cell types and molecular modalities. These results provide mechanistic insights into COVID-19 severity beyond what jointly shared variation alone could reveal.

The DIVAS package is designed to be accessible to researchers without specialised expertise in matrix perturbation theory or convex optimisation, providing straightforward functions for decomposition and downstream interpretation. As multiomics studies become increasingly common and complex in biomedical research, DIVAS offers researchers a principled approach to extract the full spectrum of biological information encoded across complementary molecular measurements, ultimately advancing our understanding of complex biological systems and disease mechanisms.

Author contributions JM and JSM conceived the project; JM and KALC secured funding; YS developed the R package and tested it on simulated and real-world datasets under the supervision of JM; JM and YS performed data analyses for the COVID-19 case study and compiled the vignette; JM and YS wrote the manuscript; all authors participated in revising the manuscript.

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Data availability The raw scRNA-seq count tables for the COVID-19 case study can be accessed by Array Express under the accession number E-MTAB-9357; metabolomic and proteomic datasets are available from Mendeley Data [40]. Processed data used in this paper

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are available from Zenodo [41].

Code availability The DIVAS R package is available on GitHub [18]. The pipeline used for processing COVID-19 data is available on GitHub [42]. The code for reproducing results in this paper is available on GitHub [43].

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Competing interests The authors declare no competing interests.

Use of large language model (LLM) For generating the summary of metabolic and proteomic pathways in Fig 2D, Claude Sonnet 4.5 was used to identify potential pathways based on features with top DIVAS loading values. These pathways were then validated by a though literature search and review.

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